

Changyong Kwak

Madison, WI | (608) 515-0546 | ckwak7@wisc.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Wisconsin-Madison
Bachelor of Science in Computer Science

Expected December 2027
Madison, WI

EXPERIENCE

Software Engineer Intern
Contoro Robotics

Feb 2026 - Jul 2026
Austin, TX

- Shipped a full-stack control platform for **3 Isaac Sim robots** by building a React/TypeScript frontend on a **13-endpoint FastAPI backend**, giving **100+ users** browser access without local GPU or ROS setup.
- Orchestrated command sequencing across **4 production adapter modes** by building per-robot queues, session locks, cancellation, and timeout recovery over a ROS 2 WebSocket bridge under concurrent execution.
- Validated control-service reliability with **27 pytest cases** covering malformed input, cancellation, concurrency, and timeouts, then automated Python 3.10 to 3.12 tests, lint, type, and packaging gates in CI.

AI Research & Software Engineering Intern

Jun 2026 - Aug 2026

Summer of AI Laboratory (SAIL) | N+1 Institute @ UW-Madison

Madison, WI

- Engineered a data-driven agent-evaluation platform spanning **528 cases across 11 task families** by building typed Python contracts, deterministic call repair, and strict graders for **3 versioned domain packs**.
- Co-developed a browser-based Agent Lab that streams plans, model output, tool calls, and verifier verdicts over Server-Sent Events, giving engineers a live view of workspace state and generated artifacts.
- Hardened concurrent writes and crash recovery with writer locks, hash-bound manifests, and exact resume, verified by **461 Windows ARM64 tests** and GitHub Actions across Python 3.9 to 3.13.

Undergraduate Research Intern

May 2026 - Present

CATS Lab @ UW-Madison

Madison, WI

- Modeled traffic-camera calibration drift across **4 error sources** by analyzing intrinsic and extrinsic parameters, homographies, reprojection error, and temporal drift in roadside video sequences.
- Reproduced a **420K-parameter PyTorch Transformer** on **97,499 observations from 82 trajectories**, using trajectory-level splits and robustness checks to prevent leakage and expose session-label confounding.

Creator & Full-Stack Developer

Oct 2024 - Feb 2026

GoViral | AI Video Automation Platform

Madison, WI

- Launched a cloud-hosted video platform for **40+ beta users** and cut turnaround from **2 hours to under 10 minutes** by shipping Python/Flask APIs and parallel FFmpeg data-processing pipelines on Google Cloud Run.
- Instrumented API calls, latency, and pipeline failures in Google Cloud Monitoring across **40+ users**, isolating breaks in FFmpeg rendering, ElevenLabs voiceovers, and AssemblyAI captions in production.

PROJECTS

LiveClaw | [GitHub](#) | *Rust, Tauri, Python, macOS APIs*

Mar 2026

- Designed a fail-safe desktop control loop across **3 control layers** by integrating a Rust/Tauri HUD, Python orchestration, and macOS input APIs behind **5 safety mechanisms**, including emergency stop.
- Implemented closed-loop action verification through a **2-stage before-and-after screenshot diff**, routing screen capture and system state into planner and verifier modules that confirm each executed action.

ClearMove | [GitHub](#) | *Next.js, TypeScript, PostgreSQL, Jest*

Nov 2025

- Created a Next.js and PostgreSQL rental-inspection application that placed **3rd of 92 teams** by parsing leases into structured JSON and generating timestamped evidence PDFs for inspection audits.

TECHNICAL SKILLS

Languages: Rust, Go, TypeScript, Python, JavaScript, C/C++, SQL, Java, Bash

Full-Stack & Data: React, Next.js, Tauri, FastAPI, Flask, Node.js, REST APIs, WebSockets, Server-Sent Events, PostgreSQL, Drizzle ORM, JSON serialization, data-processing pipelines, visualization, PyTorch

Systems & Tools: memory, performance, concurrency, debugging, distributed systems, ROS 2, NVIDIA Isaac Sim (3D simulation), Linux, AWS EC2/EKS, Docker, GitHub Actions CI/CD, Git, pytest, Jest